

LN	Status	Topic
L1		Introduction to course; What is an Algorithm?
L2		Fundamentals of Algorithmic Problem Solving, Important Problem Types
L3		Fundamental Data Structures, Analysis Framework
L5		Asymptotic Notations and Basic Efficiency Classes
L6		Mathematical Analysis of Non-recursive Algorithms & Recursive Algorithms
L7		Selection Sort and Bubble Sort
L9		Sequential Search, Brute-Force String Matching
L10		Exhaustive Search
L11		Depth First Search (DFS), Breadth First Search (BFS)
L13		Insertion Sort
L14		Topological Sorting
L15		Binary Search
L17		Merge Sort
L18		Quick Sort
L19		Binary tree traversals and related properties, Multiplication of large integers
L21		Stassen's Matrix Multiplication
L22		Presorting, Balanced Search Trees – AVL trees
L23		Balanced Search Trees – 2-3 trees
L25		Heaps and Heapsort
L26		Problem Reduction, Sorting by Counting
L27		Input Enhancement in String Matching – Horspool algorithm
L29		Boyer-Moore Algorithm
L30		Hashing
L31		Computing a Binomial Coefficient, Warshall's algorithm
L33		Floyd's Algorithm
L34		Knapsack Problem Bottom-up
L35		Knapsack-Memory Functions
L37		Prim's Algorithm
L38		Kruskal's Algorithm
L39		Dijkstra's Algorithm
L41		Huffman Trees
L42		Backtracking: n – Queen's problem
L43		Backtracking: Hamiltonian Circuit Problem, Subset-Sum Problem
L45		Branch-and-Bound: Assignment Problem, Knapsack Problem
L46		Traveling Salesperson
L47		P, NP and NP Complete Problems: Definition